Q4: Torsion Zeros and Transverse Weyl Curvature

Local structure, positive transverse indices, and the remaining global issue

Research continuation from the supplied four-dimensional Einstein manuscripts

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Abstract

For an almost-Kähler Einstein four-manifold we derive an exact second-derivative identity at every torsion zero. Combining this identity with the adapted-frame curvature equations gives a complex normal form for the derivative of the torsion, a nonnegative real determinant, and a sharp algebraic estimate involving the transverse Weyl component. In particular, a torsion zero where that component is nonzero is isolated and transverse, with positive index. If all zeros are transverse on a closed manifold, their number is $5\chi + 6\tau$. These are necessary conditions for an actual solution, not an existence theorem for prescribed jets. They do not establish the closed negative Goldberg conclusion.

1 Conventions and the precise result

Let (M^4, g, J, Ω) be a smooth, positive-definite, oriented almost-Kähler Einstein manifold. Thus

$$d\Omega = 0, \qquad \Omega(X,Y) = g(JX,Y), \qquad |\Omega|^2 = 2, \qquad \mathrm{Ric} = \frac{s}{4}g.$$

The scalar s is constant. The local results do not require completeness, compactness, or a sign for s. Write $W = W^+$ as a symmetric trace-free endomorphism of Λ^+, with Hilbert–Schmidt operator norm. Set

$$q = |\nabla\Omega|^2, \quad n = \Omega/\sqrt{2}, \quad t = \langle Wn, n\rangle = \frac{s}{6} + \frac{q}{4}, \quad \eta = W\Omega - t\Omega, \quad v = \eta/\sqrt{2}. \tag{1}$$

With respect to $\mathbb{R}n \oplus n^\perp$,

$$W = \begin{pmatrix} t & v^* \\ v & -\frac{t}{2}\,\mathrm{Id} + V \end{pmatrix}, \qquad \mathrm{tr}\, V = 0, \qquad |W|^2 = \frac{3}{2}t^2 + |\eta|^2 + |V|^2. \tag{2}$$

Here V is the W_3^+ component in the chosen almost-Hermitian splitting, not the full restriction $W|_{n^\perp}$.

Locally choose $\phi, J\phi$ of norm $\sqrt{2}$, orthogonal to Ω. The adapted-frame equations of [1, Section 2.3] are

$$\nabla\Omega = a \otimes \phi - Ja \otimes J\phi, \qquad q = 4|a|^2, \tag{3}$$

$$da - Ja \wedge b = -\mathcal{R}(J\phi), \qquad d(Ja) + a \wedge b = -\mathcal{R}(\phi). \tag{4}$$

The covector convention is $(Ja)(X) = -a(JX)$. At a zero p of a, let

$$L : T_pM \longrightarrow T_pM, \qquad L(X) = (\nabla_X a)^\sharp.$$

This derivative is independent of derivatives of the chosen frame at p. Changing ϕ multiplies L on its target by a unitary rotation, so its real determinant, singular values, and vanishing rank are well defined.

Theorem 1.1 (Local structure of a torsion zero)**.** *At every point p where $q(p) = 0$:*

$$\boxed{|\nabla^2\Omega|^2 = 4|V|^2 - 6|\eta|^2.} \tag{5}$$

There are complex orthonormal identifications of the source and target of L with $\mathbb{C}^2$ in which

$$L(z) = cz + A\bar{z}, \qquad c \in \mathbb{C}, \quad A \in M_2(\mathbb{C}), \qquad \operatorname{tr}(A\bar{A}) = -8|c|^2. \tag{6}$$

The bar in $A\bar{A}$ is entrywise conjugation, not the Hermitian adjoint. Furthermore,

$$|\eta|^2 = 8|c|^2, \tag{7}$$

$$\boxed{|V|^2 - 4|\eta|^2 = 4\|\operatorname{Sym} A\|_F^2 \geq 0,} \tag{8}$$

$$\boxed{\det_{\mathbb{R}} L = 9|c|^4 + |\det_{\mathbb{C}} A|^2 \geq 25|c|^4 = \frac{25}{64}|\eta|^4.} \tag{9}$$

Consequently, if $\eta(p) \neq 0$, the torsion zero is isolated and transverse, and q has a nondegenerate local minimum there. Every transverse torsion zero has positive local index. If L is singular, then $\eta(p) = 0$ and $\operatorname{rank}_{\mathbb{R}} L \in \{0, 2\}$.

2 The second-derivative identity

Use the nonnegative scalar Laplacian $\Delta = -\operatorname{div}\nabla$ and the nonnegative rough Laplacian $\nabla^*\nabla$. Since Ω is a self-dual harmonic form,

$$\nabla^*\nabla\Omega = 2W\Omega - \frac{s}{3}\Omega = \frac{q}{2}\Omega + 2\eta. \tag{10}$$

This also proves the expression for t in (1). Einstein metrics satisfy $\delta W = 0$, and the classical Weyl–Weitzenböck identity is

$$\nabla^*\nabla W + \frac{s}{2}W - 6W^2 + 2|W|^2 \operatorname{Id} = 0. \tag{11}$$

See [2]; the same operator normalization appears in [3, Lemma 2.5, with $f = 1$].

At a torsion zero, $\nabla n = 0$, $t = s/6$, and $\nabla^*\nabla n = 2v$. Differentiating $t = \langle Wn, n\rangle$ therefore gives

$$\begin{aligned}
\Delta t &= \langle(\nabla^*\nabla W)n, n\rangle + 2\langle W(\nabla^*\nabla n), n\rangle \\
&= 6(t^2 + |v|^2) - \frac{s}{2}t - 2|W|^2 + 4|v|^2 \\
&= 3t^2 - \frac{s}{2}t + 6|v|^2 - 2|V|^2 \\
&= 3|\eta|^2 - 2|V|^2.
\end{aligned}$$

Since $\Delta q = 4\Delta t$, whereas $\Delta|\nabla\Omega|^2 = -2|\nabla^2\Omega|^2$ at a zero of $\nabla\Omega$, this proves (5). All derivatives here are full covariant derivatives; $\nabla^2\Omega$ is not merely its skew-symmetrization.

3 The complete first derivative and its determinant

Choose an oriented orthonormal coframe at p with

$$\Omega = e^{12} + e^{34}, \quad \phi = e^{13} - e^{24}, \quad J\phi = e^{14} + e^{23}.$$

At $a(p) = 0$, also $\nabla J(p) = 0$. Einstein curvature preserves Λ^+, so (4) shows that da and $d(Ja)$ are self-dual at p. These give six independent real linear equations on ∇a. Equating their anti-self-dual components gives the following complete ten-parameter matrix. Put

$$c = \ell + im, \qquad A = \begin{pmatrix} p + iq_0 & r + it_0 \\ u + iv_0 & w + iz_0 \end{pmatrix}.$$

The matrix with derivative index on rows, namely L^{T}, is

$$\begin{pmatrix} \ell + p & m + q_0 & u & v_0 \\ -m + q_0 & \ell - p & v_0 & -u \\ r & t_0 & \ell + w & m + z_0 \\ t_0 & -r & -m + z_0 & \ell - w \end{pmatrix}. \tag{12}$$

Its transpose is exactly the real matrix of $z \mapsto cz + A\bar{z}$. Thus no additional modes have been discarded.

Write $\mathcal{R}^+ = W + s\,\mathrm{Id}\,/12$. At the zero, the transverse 2×2 block of $\mathcal{R}^+$ is V, because $t = s/6$. From $\mathcal{R}(\phi) = -d(Ja)$ and $\mathcal{R}(J\phi) = -da$ one obtains

$$(\mathcal{R}^+)_{12} = -2\ell, \qquad (\mathcal{R}^+)_{13} = -2m, \qquad V = \begin{pmatrix} v_0 - t_0 & r - u \\ r - u & t_0 - v_0 \end{pmatrix}.$$

Hence

$$|\eta|^2 = 8|c|^2, \qquad |V|^2 = 2|A_{12} - A_{21}|^2, \qquad |L|^2 = 4|c|^2 + 2\|A\|_F^2. \tag{13}$$

Differentiating (3) at $a = 0$ gives $|\nabla^2\Omega|^2 = 4|L|^2$. Thus (5) is precisely

$$8|c|^2 + \|A\|_F^2 - |A_{12} - A_{21}|^2 = 0.$$

The last two terms equal $\mathrm{tr}(A\bar{A})$, which proves (6).

Let $A = S + K$, where S is complex symmetric and K is complex skew-symmetric (that is, $S = (A + A^{\mathsf{T}})/2$). Then

$$\mathrm{tr}(A\bar{A}) = \|S\|_F^2 - \|K\|_F^2, \qquad |V|^2 = 4\|K\|_F^2.$$

Equation (6) yields $\|K\|_F^2 = \|S\|_F^2 + 8|c|^2$, and proves (8). In particular the norm identity also reads

$$|\nabla^2\Omega|^2 = 10|\eta|^2 + 16\|S\|_F^2.$$

The real determinant of a complex-linear plus anti-linear map is

$$\begin{aligned} \det_{\mathbb{R}}(c\,\mathrm{Id} + A\mathsf{C}) &= \det_{\mathbb{C}} \begin{pmatrix} c\,\mathrm{Id} & A \\ \bar{A} & \bar{c}\,\mathrm{Id} \end{pmatrix} \\ &= |c|^4 - |c|^2\,\mathrm{tr}(A\bar{A}) + |\det A|^2, \end{aligned}$$

where C is complex conjugation. The equality holds for all c, by polynomial continuation if necessary. Substitution proves the first equality in (9).

For $A = \left(\begin{smallmatrix} a & b \\ d & e \end{smallmatrix}\right)$,

$$\mathrm{tr}(A\bar{A}) = |a|^2 + |e|^2 + 2\Re(b\bar{d}) \geq 2|ae| - 2|bd| \geq -2|ae - bd|.$$

It follows that $|\det A| \geq 4|c|^2$, which proves the lower bound in (9). Equality in both (8) and (9) is possible for the algebraic data

$$c \in \mathbb{R}, \qquad A = \begin{pmatrix} 0 & 2c \\ -2c & 0 \end{pmatrix}.$$

This establishes sharpness of the algebraic coefficients under the necessary equations above; it does not construct an Einstein metric realizing these prescribed data.

If $\eta \neq 0$, the determinant is positive. The inverse function theorem applied to a gives an isolated transverse zero; since $q = 4|a|^2$,

$$\mathrm{Hess}_p\, q(X, Y) = 8\langle LX, LY\rangle.$$

If $\det L = 0$, (9) forces $c = 0$ and $\det A = 0$. Then $L = A\mathsf{C}$ has real rank zero or two. This completes the local algebraic assertions of Theorem 1.1.

4 Indices on a closed manifold

Use the standard Nijenhuis convention

$$N_J(X, Y) = [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y].$$

The adapted-frame expression for ∇J implies

$$N_J(e_1, e_3) = 2Ja^\sharp.$$

After complexifying this is an orientation-preserving linear identification of the four real torsion components with the complex rank-two bundle

$$\mathcal{N} = \Lambda^{0,2}T^*M \otimes T^{1,0}M.$$

Changing the adapted frame changes this identification by orientation-preserving complex transformations. At a zero, derivatives of the identification itself multiply $a = 0$ and do not affect the Jacobian. Thus the sign of a transverse local Nijenhuis index is the sign of $\det L$. By (9), it is positive.

Corollary 4.1 (Count when the torsion zeros are transverse)**.** *If M is closed and every zero of N_J is transverse, then*

$$\boxed{\#\{N_J = 0\} = 5\chi(M) + 6\tau(M).} \tag{14}$$

In particular the same formula holds if η is nonzero at every torsion zero. If a torsion zero is degenerate, then η vanishes there and the first derivative has real rank zero or two.

Proof. The zeros are isolated and finite. If x_1, x_2 are the formal Chern roots of $T^{1,0}M$, those of $\mathcal{N}$ are $2x_1 + x_2$ and $x_1 + 2x_2$. Therefore

$$c_2(\mathcal{N}) = 2c_1^2(TM, J) + c_2(TM, J) = 5\chi + 6\tau.$$

The index theorem for a transverse section equates this Chern number to the sum of local indices. Each is $+1$ by Theorem 1.1. The characteristic-class formula is the classical input used in the project's torsion-zero argument, following [4]. □

For a closed Einstein metric with $s \neq 0$, Chern–Weil gives $\chi > 0$ and $2\chi + 3\tau > 0$, so $5\chi + 6\tau = \chi + 2(2\chi + 3\tau) > 0$. Consequently (14) is compatible with Einstein topology. It is not a contradiction. Transversality is an extra property of the actual Nijenhuis section and is not obtained by perturbing that section while discarding the almost-Kähler Einstein equations.

5 A scalar current identity and its limit

For clarity, the pointwise identity behind the usual integrated balance can also be retained without dropping its current. Define the one-form

$$\theta(X) = \langle \eta, \nabla_X \Omega \rangle.$$

Then the same Einstein and harmonic-form equations imply

$$\boxed{\frac{1}{4}\Delta q + 2\,\mathrm{div}(\theta^\sharp) + \frac{s}{8}q + \frac{3}{16}q^2 + |\eta|^2 + 2|V|^2 = 0.} \tag{15}$$

Here is a direct derivation. Let $T_i = \nabla_i n$, $f = \sum_i |T_i|^2 = q/2$, and $\beta_i = \langle v, T_i \rangle = \theta_i/2$. The almost-Kähler form of (3) gives

$$\sum_i T_i \otimes T_i = \frac{f}{2}(\mathrm{Id} - n \otimes n), \quad \sum_i \langle W T_i, T_i \rangle = -\frac{tf}{2}, \quad \nabla^* \nabla n = fn + 2v.$$

Product differentiation using (11) first yields

$$\Delta t = 3t^2 - \frac{s}{2}t + 6|v|^2 - 2|V|^2 + 3tf - 4\sum_i \langle (\nabla_i W) T_i, n \rangle.$$

A separate differentiation of β gives

$$\sum_i \langle (\nabla_i W) T_i, n \rangle = \mathrm{div}\,\beta + \frac{3}{2}tf + 2|v|^2.$$

Since $f = 2t - s/3$, their combination is

$$\Delta t = \frac{s}{2}t - 3t^2 - 2|v|^2 - 2|V|^2 - 4\,\mathrm{div}\,\beta.$$

Substitution of (1) proves (15).

On a closed manifold, integration recovers the balance already contained in the supplied manuscript:

$$\int \left(|\eta|^2 + 2|V|^2 + \frac{3q^2}{16} \right) dV_g = -\frac{s}{8} \int q\, dV_g.$$

That integrated identity is not an additional constraint introduced here. At a torsion zero, $\theta = 0$ but $\mathrm{div}\,\theta = -2|\eta|^2$, so dropping the divergence pointwise would be incorrect. Keeping it recovers (5).

Connection to Q4. The supplied manuscript previously established that a closed nonzero-scalar Einstein almost-Kähler candidate has a torsion zero. The result here specifies the full first derivative at such a zero: nonzero transverse Weyl curvature forces a positive isolated zero, whereas a degenerate zero must lie in the locus $\eta = 0$ and has even derivative rank. None of these statements proves that η vanishes on all of M. The compact third-Gray theorem in [1] would finish the integrability argument if that global vanishing were obtained. A pointwise zero of the torsion is not, by itself, the infinite-order vanishing required by a strong unique-continuation argument.

Reproducible finite calculations. The accompanying `verify_zero_structure.py` checks the six linear self-duality conditions, their ten-dimensional solution space, the norm identities, the determinant formula, and the algebraic equality case. It also substitutes the project's explicit rational noncompact metric into (15); the rational residual is exactly zero. That is a consistency check on a previously supplied model, not a new example or a metric-existence argument for the algebraic data at a torsion zero.

References

[1] V. Apostolov, J. Armstrong and T. Drăghici, *Local models and integrability of certain almost Kähler 4-manifolds*, Math. Ann. **323** (2002), 633–666. arXiv:math/0109098. Equations (21)–(24), the harmonic-form identities, and Theorem 2.

[2] A. Derdziński, *Self-dual Kähler manifolds and Einstein manifolds of dimension four*, Compositio Math. **49** (1983), 405–433. Journal archive.

[3] G. Catino and D. Dameno, *Conformal Kähler Rigidity of Einstein Four-Manifolds*, 2026 preprint, Lemma 2.5. arXiv:2607.21538. Only the classical Weyl–Weitzenböck formula, with $f = 1$, is used here.

[4] J. Armstrong, *Almost-Kähler Geometry*, doctoral thesis. Author's text. Characteristic class of the Nijenhuis bundle.

[5] Supplied project manuscripts, 8 October 2026: `01_Research_Synthesis(1).tex`, the Goldberg section; `02_Mathematical_Source_Companion(1).tex`, Step 55; and `03_Subsequent_Mathematical_Developments(1).tex`, the explicit Einstein example and the closed torsion-zero proposition. These identify the existing project results to which the local calculation is connected. No FD, N1, or global Einstein-gap theorem is used in this proof.